Lecture 10

Optimal Policy Learning, Text-as-Data & Images

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	KEY QUESTION
1	Optimal Policy Learning	Who should receive treatment?
2	Text as Data	How do we turn language into numbers?
3	Images as Data	What can pixels tell economists?

Goal: Complete the pipeline: prediction → causal inference → **policy decisions**

Part 1

Optimal Policy Learning

From Estimation to Decision

The journey so far:

STEP	TOOL	OUTPUT
Prediction	ML models	$\hat{y}$
Identification	DAGs, CIA, IV	Causal structure
Estimation	DML, Causal Forests	$\hat{\tau}(x)$
Decision	**Policy Learning**	**Who to treat?**

The missing piece: knowing $\tau(x)$ is not enough — we need a *policy rule* that respects budget constraints.

The Policy Problem

Setting: Job Training Program *(illustrative example for teaching)*

- Government budget: train 1,000 out of 10,000 workers
- Cost per person: $5,000
- Wage gains vary: $\tau(x_i) \in [-\$2{,}000,\ \$15{,}000]$

Numbers are hypothetical, designed to illustrate budget-constrained policy learning.

Two wrong answers:

- ❌ Treat everyone with positive $\hat{\tau}$ → budget exceeded
- ❌ Target the highest average effect group → ignores individual variation

Right answer: Learn a *policy rule* $\pi : \mathcal{X} \to \{0, 1\}$ that maximizes total welfare under the budget.

The Policy Learning Framework

Objective: Maximize expected welfare

$$V(\pi) = E\left[\pi(X) \cdot \left(Y(1) - c\right) + (1 - \pi(X)) \cdot Y(0)\right]$$

Optimal rule (if $\tau(x)$ were known):

$$\pi^*(x) = \mathbf{1}\{\tau(x) > c\}$$

The problem: $\tau(x)$ is estimated from data with statistical error. Plugging in $\hat{\tau}(x)$ directly ignores uncertainty and can lead to poor out-of-sample performance.

Key idea (Athey & Wager 2021): Directly optimize policy on observed data using *doubly robust welfare scores*.

Empirical Welfare Maximization (EWM)

Kitagawa & Tetenov (2018): The policy learning objective is a sample analogue of expected welfare.

$$\hat{W}_n(\pi) = \frac{1}{n}\sum_{i=1}^{n} \hat{\Gamma}_i \cdot \pi(X_i)$$

The EWM estimator:

$$\hat{\pi}_{EWM} = \arg\max_{\pi \in \Pi} \hat{W}_n(\pi)$$

Why this works: As $n \to \infty$, $\hat{W}_n(\pi) \to W(\pi)$ uniformly over policy class Π — if Π is not too complex (VC dimension bounded).

Regret bound: $W(\pi^*) - W(\hat{\pi}_{EWM}) = O_p(n^{-1/2})$ for finite policy classes.

From CATE to Policy

Knowing $\hat{\tau}(x)$ is not enough.

PROBLEM	WHY $\hat{\tau}(x)$ FAILS	WHAT POLICY LEARNING ADDS
Budget	Treating all $\tau > 0$ exceeds capacity	Top-k selection under constraint
Fairness	Subgroup disparities in treatment	Constrain π to exclude protected attributes
Robustness	Overfitting to estimated scores	DR scores reduce first-stage bias
Interpretability	Black-box CATE hard to defend	Decision tree rules are auditable

> *Policy learning = CATE estimation + constrained optimization + welfare evaluation.*

Doubly Robust Welfare Scores

Replace unobserved $\tau(x_i)$ with the score:

$$\hat{\Gamma}_i = \hat{\tau}(X_i) + \frac{W_i\big(Y_i - \hat{\mu}_1(X_i)\big)}{\hat{e}(X_i)} - \frac{(1 - W_i)\big(Y_i - \hat{\mu}_0(X_i)\big)}{1 - \hat{e}(X_i)}$$

Three parts: (1) model-based imputation $\hat{\tau}(X_i)$; (2) IPW correction for treated; (3) IPW correction for controls.

Why doubly robust? $\hat{\Gamma}_i$ is unbiased if *either* the outcome model $\hat{\mu}$ or the propensity score $\hat{e}$ is correctly specified — not both required.

Result: Policy learning problem becomes weighted classification:

$$\max_{\pi \in \Pi} \frac{1}{n} \sum_{i=1}^{n} \hat{\Gamma}_i \cdot \pi(X_i)$$

Policy Trees

Simple, interpretable policy rules:

```
if Age < 30 and Education ≥ 12:
    → Treat  (young, educated workers benefit most)
else:
    → Don't treat
```

Why trees?

- Interpretable by policymakers and courts
- Partitions feature space into treatment/control regions
- Can impose fairness constraints (no race, gender)

Depth d controls complexity: depth-2 gives 4 leaves, depth-3 gives 8 leaves

Budget Constraints

With fixed budget (treat at most k individuals):

$$\max_{\pi} \sum_{i=1}^{n} \hat{\Gamma}_i \pi(X_i) \quad \text{s.t.} \quad \sum_{i=1}^{n} \pi(X_i) \leq k$$

Solution: Top-k rule

1. Compute $\hat{\Gamma}_i$ for all individuals
2. Rank by $\hat{\Gamma}_i$ descending
3. Treat top k

Policy tree with budget: Enforce budget as a constraint on the tree — each leaf has a maximum allocation.

Numerical Illustration: Medicare-Style Plan Assignment

Setting: Which insurance plan should each consumer enroll in to minimize out-of-pocket costs?

Policy learning approach: Train a decision tree on doubly robust welfare scores; assign consumers to plans using only observable characteristics (age, health status, prior utilization).

	NAIVE ASSIGNMENT	POLICY TREE (DEPTH 3)	THEORETICAL OPTIMAL
Annual savings per person	$0	**$270**	$300
% of optimal achieved	—	**90%**	100%

Values are illustrative, based on stylized simulations in the spirit of Athey & Wager (2021) and Kitagawa & Tetenov (2018). A depth-3 tree (8 leaves) approximates the optimal rule with simple, auditable criteria.

Real-world analog: Abaluck & Gruber (2011) show Medicare Part D enrollees often choose dominated plans; policy trees offer a scalable, interpretable assignment mechanism.

Fairness in Policy Learning

Problem: Maximizing welfare may allocate treatment unfairly across demographic groups.

Fairness constraints:

- **Demographic parity:** $P(\pi(X) = 1 \mid G = g)$ equal across groups g
- **Equalized opportunity:** False negative rates equal across groups
- **No protected attributes in π:** Exclude race, gender from policy rule

Trade-off: Fairer policies usually achieve lower total welfare. The frontier is estimable via constrained optimization.

Reference: Kitagawa & Tetenov (2018) provide welfare bounds under fairness constraints.

Welfare Regret and Sample Complexity

Welfare regret: $R(\hat{\pi}) = V(\pi^*) - V(\hat{\pi})$ — how much worse is $\hat{\pi}$ vs the optimal rule?

Athey & Wager (2021): For a depth-d tree over p features:

$$R(\hat{\pi}) = O_p\left(\sqrt{\frac{2^d \log p}{n}}\right)$$

DEPTH	LEAVES	RATE
2	4	$O_p(n^{-1/2}\sqrt{\log p})$ — fast
3	8	$O_p(n^{-1/2}\sqrt{\log p})$ — fast
Large	—	Much slower — overfitting

Why DR scores help: Lower score variance → smaller bound constant → faster convergence.

Implication: Depth-3 trees suffice for $n \gtrsim 1{,}000$.

Multi-Arm Policy Learning

Binary treatment: $\pi(X) \in \{0, 1\}$ — treat or not.

Multiple arms: $\pi(X) \in \{0, 1, \ldots, K\}$ — which of $K+1$ options?

Example: Job training — *no training*, *short course*, *long course*.

DR welfare score for arm k:

$$\hat{\Gamma}_i^{(k)} = \hat{\tau}_k(X_i) + \frac{\mathbf{1}(W_i = k)\big(Y_i - \hat{\mu}_k(X_i)\big)}{\hat{e}_k(X_i)}$$

Assignment rule: Send each unit to the arm with the highest DR score:

$$\hat{\pi}(X_i) = \arg\max_k \hat{\Gamma}_i^{(k)}$$

Multi-arm policy tree: Each leaf selects the arm with the highest average DR score.

Policy Learning: Packages

PACKAGE	LANGUAGE	KEY FUNCTION
`econml`	Python	`DRPolicyTree`, `DRPolicyForest`
`policytree`	R	`policy_tree()`
`grf`	R	`multi_arm_causal_forest()`

Workflow: GRF or DRLearner for $\hat{\tau}(x)$ → compute $\hat{\Gamma}_i$ → fit policy tree

Implementation: `Lecture10-Exercise.ipynb` — Part 1 implements DRPolicyTree on a job training dataset.

Part 1 Summary: Optimal Policy Learning

CONCEPT	WHAT IT DOES	KEY REFERENCE
DR welfare score	Unbiased estimate of individual treatment effect	Athey & Wager (2021)
Policy tree	Interpretable decision rule	Kitagawa & Tetenov (2018)
Budget constraint	Treat only top k by score	Top-k rule
Fairness	Equal allocation across groups	Constrained EWM
Regret bound	Statistical guarantee on welfare	$O_p(\sqrt{2^d \log p/n})$

> ***Workflow:*** *Estimate CATE → Compute DR scores → Learn policy tree → Evaluate via cross-fitting.*

Part 2

Text as Data

The Opportunity: Unstructured Economic Data

Traditional economics: Prices, quantities, income, survey responses — all numbers.

The new frontier:

- FOMC minutes → monetary policy stance
- Earnings call transcripts → guidance sentiment
- Job postings → skill demand and wage premia
- Congressional speeches → political polarization index

The challenge: Computers see text as sequences of characters. We need text → vectors that preserve *meaning*.

Text Preprocessing: Before Any Embedding

Raw text must be cleaned before vectorisation. Steps differ by method.

STEP	WHAT IT DOES	EXAMPLE
Tokenise	Split into words / subwords	`"central bank."` → `["central", "bank"]`
Lowercase	Remove case variation	`"Federal"` → `"federal"`
Remove stops	Drop high-frequency noise words	`["the", "rate"]` → `["rate"]`
Stem / lemmatise	Reduce inflections	`"raising"` → `"raise"`

When to skip which step:

- **TF-IDF:** apply all steps — smaller vocabulary, better discrimination
- **Word2Vec:** remove stops optional; keep domain vocabulary intact
- **BERT / SBERT:** skip *all* steps — full sentence context is required

Three Layers of Text Representation

LAYER	METHOD	CAPTURES	SPEED
Frequency	BoW / TF-IDF	What words appear?	⚡⚡⚡
Semantics	Word2Vec	What do words mean?	⚡⚡
Context	BERT / SBERT	How does meaning shift?	⚡

Start simple. Upgrade only when needed.

Layer 1: TF-IDF

Term Frequency × Inverse Document Frequency:

$$\text{TF-IDF}_{t,d} = f_{t,d} \times \log \frac{N}{|\{d : w_t \in d\}|}$$

- **TF**: how often word t appears in document d
- **IDF**: penalizes words common across all documents ("the", "is")
- **Result**: words that are distinctive to a document get high weight

Gentzkow & Shapiro (2010): Measured newspaper political bias by TF-IDF similarity to Democrat vs Republican Congressional speech — no human labeling needed.

Layer 2: Word Embeddings (Word2Vec)

Problem with TF-IDF: "car" and "automobile" → completely different dimensions — no semantic link.

Word2Vec (Mikolov et al. 2013): Train a neural net to predict surrounding words. Words with similar contexts get similar vectors.

$$"\text{King}" - "\text{Man}" + "\text{Woman}" \approx "\text{Queen}"$$

BERT (Devlin et al. 2019): Context-sensitive.

- "The *bank* is steep" → river bank embedding
- "The *bank* raised rates" → financial bank embedding

Same word, different meaning, **different vector**.

Pre-training vs Fine-tuning

BERT training pipeline:

1. **Pre-training** (unsupervised, large corpus):
   - Masked language modeling: predict missing words
   - Next sentence prediction: does sentence B follow A?
   - Cost: ~$10k$ GPU-hours on Wikipedia + BookCorpus
2. **Fine-tuning** (supervised, small labeled data):
   - Add classification head on top of [CLS] token
   - Train only the head (or head + top layers)
   - Cost: minutes to hours on a single GPU

Economics implication: Fine-tune BERT on financial/economic text (10-K, FOMC) for domain-specific embeddings.

BERT: The Attention Mechanism

Why attention? Word meaning depends on *all* surrounding words, not just nearby ones.

Scaled dot-product attention (Vaswani et al. 2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Q (query), K (key), V (value) are linear projections of each token. The softmax assigns attention *weights* across all other tokens.

What attention learns for "bank":

SENTENCE	ATTENDS MOST TO
"The *bank* raised rates"	raised, rates, monetary
"The *bank* of the river"	river, steep, water

Context-sensitive embeddings — the same token produces different vectors in different sentences.

Layer 3: Sentence Embeddings (SBERT)

Problem with BERT: Word-level only; expensive to compare documents.

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (Reimers & Gurevych 2019):

- One embedding per sentence/document
- Cosine similarity directly measures semantic distance
- Fed vs ECB vs PBOC statements: track policy divergence over time

Economics application: Roberts & Wheaton (2023) — pairwise SBERT similarity across central bank statements reveals that policy synchronization *decreases* during crises.

Application: Text as Confounder in DML

Text creates confounding — and DML can fix it.

Example: Does a product promotion increase sales?

- Confounder: product review sentiment (affects both promotion decisions *and* sales)
- Naive OLS: biased — conflates promotion effect with review quality

DML solution:

1. Convert reviews → TF-IDF features (high-dimensional controls W)
2. Partial out text effects: regress Y and D on W using Lasso
3. Recover text-adjusted causal effect from residuals

This directly connects Part 1 (DML) with text data.

Text as Data: Summary

LAYER	METHOD	BEST FOR
Frequency	TF-IDF	Classification, fast prototyping
Semantics	Word2Vec	Similarity, analogy, word-level
Context	BERT	Nuanced sentiment, NER, QA
Sentence	SBERT	Document similarity, clustering

Reference: Gentzkow, Kelly & Taddy (2019, *JEL*) — *Text as Data*

Implementation: `Lecture10-Exercise.ipynb` — Part 2 covers TF-IDF, sentiment classification, and SBERT embeddings.

LLMs in Economics

Beyond embeddings: LLMs can *reason* about economic text — no fine-tuning required.

TASK	CLASSIC NLP	LLM (ZERO-SHOT)
Sentiment	Train classifier	"Is this hawkish or dovish?"
Entity extraction	NER model	"List all firms mentioned"
Survey simulation	Survey data	LLM as synthetic respondent
Summarisation	Extractive	Abstractive reasoning

Research uses: FOMC hawk/dove signals; 10-K risk factors; central bank comparisons

Caution: Always validate LLM outputs — they hallucinate plausible-sounding facts.

Part 3

Images as Data

What Satellites and Cameras See

Economists traditionally rely on surveys and administrative records. Images bypass both.

SOURCE	WHAT WE MEASURE	SURVEY ALTERNATIVE
Satellite (nightlights)	Economic activity, poverty	Household surveys
Satellite (agriculture)	Crop yields, drought	Crop reports
Street View	Housing quality, gentrification	Hedonic regressions
Port cameras	Shipping container counts	Trade statistics

Common thread: Images are cheap, frequent, and spatially granular in ways surveys cannot be.

Convolutional Neural Networks (CNNs)

Core idea: Learn a hierarchy of spatial features from raw pixels.

```
Input (224×224×3)
→ Conv + ReLU    (edges, lines)
→ Max Pooling    (reduce size)
→ Conv + ReLU    (textures, shapes)
→ Max Pooling
→ Fully Connected  →  prediction
```

Early layers: edges. Middle layers: textures. Deep layers: objects.

Image Preprocessing for Economics

Raw satellite images are not ready for ML:

PROBLEM	SOLUTION	TOOL
Cloud cover	Temporal compositing (median pixel)	Google Earth Engine
Seasonal variation	Same-month comparisons	Date filtering
Different resolutions	Resample to common grid	`rasterio`
Atmospheric distortion	Surface reflectance correction	`sen2cor`
Georeferencing mismatch	Coregistration to base layer	`GDAL`

Standard pipeline:

`Raw image → Cloud mask → Median composite → Resample → Normalize → CNN input`

Vision Transformers (ViT): The Modern Alternative

CNNs: Strong at local features (edges, textures). But share weights globally — may miss long-range spatial dependencies.

ViT (Dosovitskiy et al. 2020): Apply transformer attention directly to image patches.

```
Image (224×224)  →  196 patches (16×16)
→ Linear projection  →  patch embeddings
→ Transformer encoder  (self-attention)
→ [CLS] token  →  512-dim representation
```

Why it matters for economics:

- Captures global spatial context: buildings, roads, settlement patterns
- Pre-trained ViT-B/16 outperforms ResNet-50 on most remote sensing benchmarks
- 512-dim [CLS] embedding → plug into OLS, logit, or causal forest

Jean et al. (2016): Poverty from Space

"Combining Satellite Imagery and Machine Learning to Predict Poverty"
Science, 2016

Research question: Can satellite imagery replace costly household surveys for poverty measurement in data-scarce regions?

ML: Two-step CNN transfer learning

1. Train CNN to predict *nighttime light intensity* from daytime images (abundant labels)
2. Transfer CNN features to predict *consumption expenditure* (scarce ground truth)

Why it works: Features predicting nighttime lights (roads, rooftops, fields) correlate strongly with income.

Result: Consumption r^2 ranges from 0.60 to 0.75 across five African countries — competitive with ground surveys at near-zero marginal cost.

Remote Sensing: Technical Details

Why transfer learning works for satellite imagery:

LAYER	LEARNS	TRANSFERABLE?
Early (1-3)	Edges, colors, textures	✅ Universal
Middle (4-8)	Roofs, roads, vegetation	✅ Mostly universal
Deep (9+)	Settlement patterns, density	⚠️ Domain-specific

Fine-tuning strategy: Freeze early layers, retrain deep layers on local ground truth.

Resolution trade-off:

- 30m (Landsat): free, global, low detail → aggregate measures
- 0.5m (commercial): expensive, high detail → building-level analysis

Naik et al. (2017): Measuring Urban Change

"Computer Vision Uncovers Predictors of Physical Urban Change"
PNAS, 2017

Research question: Can the visual appearance of city streets predict neighborhood economic trajectory?

Data: Google Street View images, 2007 and 2014, across US cities

ML: Extract CNN features from street photos → predict income and housing price changes → identify visual predictors of gentrification

Key finding: Visual "safety score" predicts income growth **better than demographic variables** — images capture investment signals before they appear in census data.

Beyond Poverty: Agricultural and Environmental Applications

APPLICATION	IMAGE SOURCE	ECONOMIC QUESTION
Crop yield	Multispectral (NDVI)	Futures pricing, food security
Deforestation	Monthly Landsat	Carbon credits, policy enforcement
Air quality	Sentinel-5P aerosol	Health effects, property values
Port activity	SAR radar	Supply chain disruption, trade flows
Informal housing	High-res commercial	Urban growth, inequality

Burke et al. (2021): Fine-resolution satellite data reveals within-country economic variation that household surveys often understate — particularly in Sub-Saharan Africa.

Key challenge — domain shift: Models trained in rich countries often fail in poor-country settings (different architecture, vegetation, lighting). Cross-domain transfer learning is an active research area.

Images as Data: Summary

	TEXT	IMAGES
Representation	TF-IDF, Word2Vec, BERT	CNN features + transfer learning
Key challenge	Meaning varies by context	Scale, lighting, angle variation
Economics use	Sentiment, polarization, NLP	Poverty, urban change, agriculture
Labeled data	Low — unsupervised embeddings	High — transfer learning helps

Common theme: Convert unstructured data → numeric features → standard econometric models.

Further Reading: Optimal Policy Learning

PAPER	AUTHORS	VENUE
Policy Learning with Observational Data	Athey & Wager (2021)	*Econometrica*
Who Should Be Treated? Empirical Welfare Maximization Methods for Treatment Choice	Kitagawa & Tetenov (2018)	*Econometrica*
Consumers' Costs and Benefits of Drug Plan Choice	Abaluck et al. (2020)	*AEA P&P*

Further Reading: Text as Data

PAPER	AUTHORS	VENUE
Text as Data	Gentzkow, Kelly, & Taddy (2019)	*JEL*
What Drives Media Slant?	Gentzkow & Shapiro (2010)	*Econometrica*
Efficient Estimation of Word Representations in Vector Space	Mikolov et al. (2013)	*ICLR*

Further Reading: Text as Data (continued)

PAPER	AUTHORS	VENUE
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	Devlin et al. (2019)	*NAACL*
Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks	Reimers & Gurevych (2019)	*EMNLP*

Further Reading: Images as Data

PAPER	AUTHORS	VENUE
Combining Satellite Imagery and Machine Learning to Predict Poverty	Jean et al. (2016)	*Science*
Computer Vision Uncovers Predictors of Physical Urban Change	Naik, Raskar, & Hidalgo (2017)	*PNAS*
Using Satellite Imagery to Understand & Promote Sustainable Development	Burke et al. (2021)	*Science*
An Image Is Worth 16×16 Words: Transformers for Image Recognition at Scale	Dosovitskiy et al. (2020)	*ICLR*

Data & Code

This lecture:

- Slides source: `_slides/Lecture10-Optimal-Policy-Learning-and-Text-as-Data-Slides.md`
- Exercises: `exercises/Lecture10-Exercise.ipynb`

Replication packages:

- Policy Learning: `econml` Python package (`DRPolicyTree`)
- Text: `sentence-transformers` (SBERT), `scikit-learn` (TF-IDF)
- Images: `timm` (ViT), `torchvision` (ResNet)

Course Complete

ECON6083: Machine Learning in Economics

Prediction → Causal Inference → Policy Decisions → Text & Images

ML is a tool for economic thinking, not a replacement for it.

Thank you for a great module.